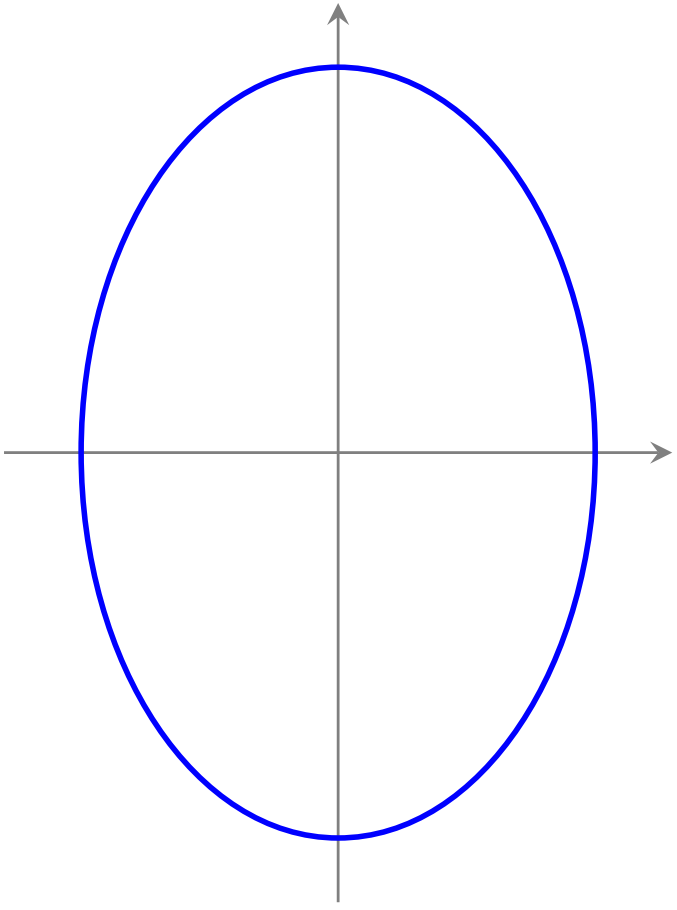
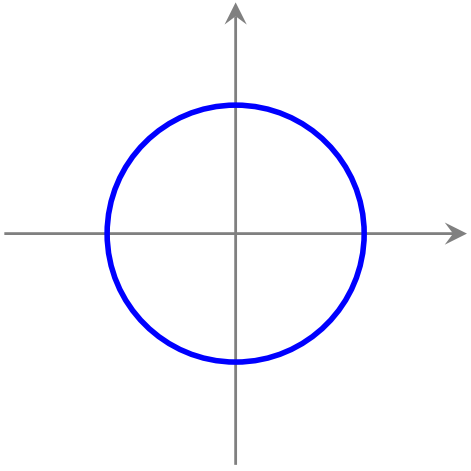


$\Sigma = \mathrm{diag}(4, 9)$

$W = \mathrm{diag}(1/2, 1/3)$



$W\Sigma W^\top = I_2$